

Mark Peng

markminpeng@gmail.com | linkedin.com/in/markminpeng | github.com/notMarkMP1 | markpeng.me

EDUCATION

University of Toronto

Toronto, Ontario, Canada

Honours B.S. in Computer Science (co-op), Minor in Math & Statistics – GPA: 3.9/4.0

Sep. 2024 – May 2028

- **Relevant Coursework:** Software Design, Systems Programming, Data Structures and Algorithms

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, TypeScript, Ruby, SQL, Lua

Frameworks: React, Node.js, Next.js, React Native, Arduino, Django, Express.js, Flask

Developer Tools: Docker, Git, GitHub, GitLab, VS Code, Jira, Postman, Linux, Copilot, Cursor

Technologies & Concepts: AWS, GraphQL, MongoDB, PostgreSQL, Redis, REST APIs, Agile, Microservices

EXPERIENCE

Block

May 2026 – Aug. 2026

Incoming Software Engineering Intern

Remote

Pulsenics

May 2025 – Aug. 2025

Full-stack Developer Intern

Toronto, Ontario, Canada

- Refactored **2K+** LOC of hardware-coupled **C firmware** into logic-based modular functions and interfaces, enabling the creation of additional **Ceedling** unit tests with mocks and stubs, raising repository test coverage by **300%**
- Implemented **10+** key features for custom **Python Modbus** client package, used by **7+** clients/internal projects
- Reduced manual verification time by **≈90%** and cut release cycle time by 30% by building a full-stack **Python QA** pipeline for post-production firmware, automating **Modbus** tests, real-time logging, and data validation
- Improved QA report load times by **10x** with custom lazy-loaded Plotly graphs in **Jinja2** built over **JavaScript**
- Minimized communication errors from **10% to 0.001%** by implementing a custom ACK protocol in **C** over Ethernet, and optimizing performance through DMA tuning and clock synchronization

Abundant Science

Feb. 2025 – Apr. 2025

Software Developer Intern

Toronto, Ontario, Canada

- Built **React Native** mobile app with camera API for PHIPA-compliant lateral flow test capture and analysis
- Engineered **≈85%** classification accuracy proof-of-concept **TensorFlow/OpenCV** model in **Python** for rapid tests
- Achieved **<3MB** bundle size and **<100ms** inference latency for on-device classification model by optimizing **TensorFlow Lite** pipeline and applying model quantization
- Integrated **GitHub Actions + Expo** for **CI/CD**, automating builds and deployments across dev and production
- Leveraged React Native Worklets with **JSI** to convert GPU pixel buffers to CPU arrays in **C++/JavaScript** for fast frame pipeline processing and rendering performance

Toronto Model United Nations

Jun. 2022 – Jul. 2024

Front-end Developer

Toronto, Ontario, Canada

- Engineered static site generation of **30+** pages using Jekyll, HTML, and SASS for modular content management

PROJECTS

🔊 reels-cli | C, Python, Bash

August 2025

- Engineered a complete terminal-based media player with a custom **C** video streaming engine and **Python** backend
- Leveraged unix domain sockets with multithreading for efficient data exchange between both **C** and **Python** processes

🔊 ProportionAI | Next.js, React, MongoDB, Terraform, Gemini

January 2025

- Developed web-app using **Next.js**, **React**, and **MongoDB** for an AI-powered time insight study app
- Deployed using **Terraform** through an **AWS EC2** deployment to host the study platform in a declarative manner

AWARDS & LEADERSHIP

Top 32 of 1000+ participants @ Hack the North, finalist for Warp – *built an AI voice agent delivery app*

Sep. 2025

UTMIST AI Open Source Developer – *team focused on contributing to AI open source projects*

May 2025

University of Toronto Scholar Award (worth \$10000), Dean's List Scholar – *for positive academic standing*

2024-2025